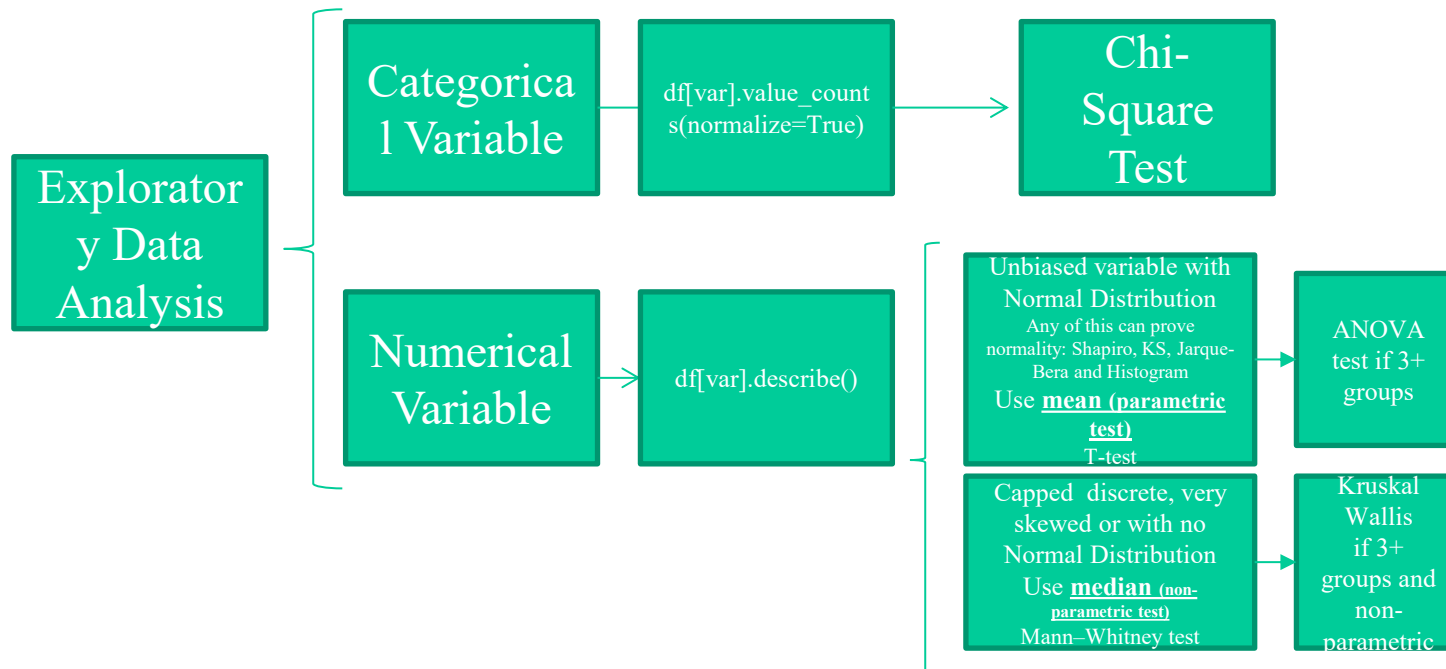




Universidad de Navarra

From Univariate Analysis to Multivariate...



- Video: Machine Learning
vs Deep Learning =

<https://www.youtube.com/watch?v=q6kJ71tEYqM>

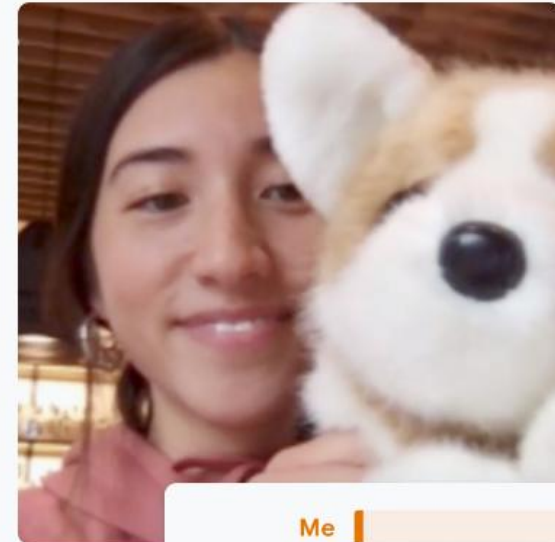
PRACTICE: No Code Example (Computer Vision Deep Learning)

Teachable Machine

Train a computer to recognize your own images, sounds, & poses.

A fast, easy way to create machine learning models for your sites, apps, and more – no expertise or coding required.

Get Started



Me

Me + Dog <3

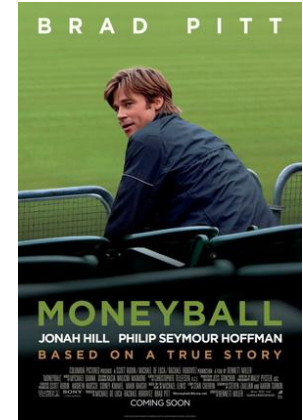
98%

<https://teachablemachine.withgoogle.com/>

- Supervised vs. Unsupervised Learning -

https://www.youtube.com/watch?v=W01tIRP_Rqs

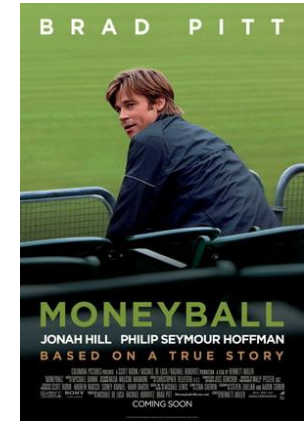
Baseball Example



➤ Moneyball -

<http://www.youtube.com/watch?v=AiAHLZVgXjk>

- Which type of company is Oakland Athletics?



EXAMPLE OF A MODEL IN USE IN A HOLLYWOOD MOVIE:

➤ Moneyball (Breaking Biases) -

<http://www.youtube.com/watch?v=yGf6LNWY9AI>

EXAMPLES OF APPLICATION OF MACHINE LEARNING

Integrated vision of the Portfolio Management approach



Main reading:

<https://mfalonso.pythonanywhere.com/static/Article-AnalyticsDrivenCompetitor-V21.pdf>

<https://www.youtube.com/watch?v=-ZuqWq25YFg>



DO YOU FEAR AN ANALYTICS DRIVEN COMPETITOR? YOU SHOULD!

AREA: IE SCHOOL OF SOCIAL AND BEHAVIORAL SCIENCES

PROFESSOR: **MANOEL F. A. GADI**

E-mail: mfalonso@faculty.ie.edu

Disruptive business models and disruptive technology are generating a frenzy in the race to have the best of breed technology and systems. The same is becoming reality in the field of statistical & computational models for Analytics. Companies like Capital One with its thousand simultaneous tests, Amazon with its recommendation systems and Google with its raking algorithm among others formed an army which took Analytics to a complete new level by making prescriptive modeling and testing the bedrock of its competitive advantage. In the

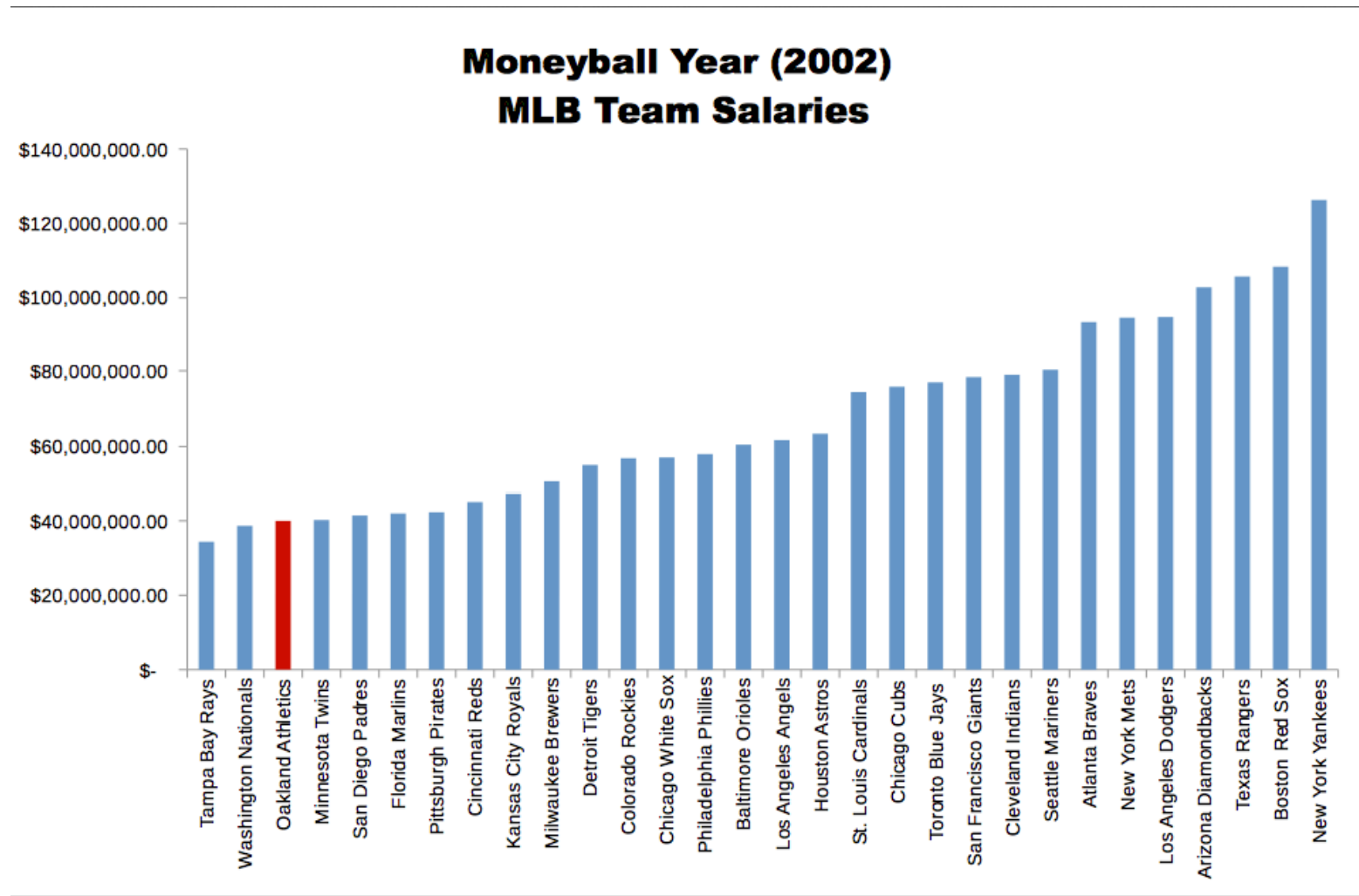
Machine Learning - Model estimation

$$f(X) = y$$



- Moneyball - <http://www.youtube.com/watch?v=AiAHlZVgXjk>

- Which type of company is Oakland Athletics?



Machine Learning - Model estimation

$$f(X) = y$$

Characteristics (select)

Target definition (another select)

f (*X*) = *y*

Population (where)

ib_var_1	ib_var_2	ib_var_3	icn_var_22	icn_var_23	icn_var_24	ico_var_25	ico_var_26	ico_var_27	if_var_65	if_var_68	if_var_80	icn_var_81
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0	0	0	4	1	1	4	4	4	50	1028	0.6666667	0
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0	1	0	2	5	3	4	3	5	39	514	0.5333334	0
0	0	0	3	1	1	4	4	4	39	1542	0.6	0
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0	0	0	3	3	1	3	4	5	55	3084	0.6	1
0	1	0	2	1	1	4	5	4	45	616.8	0.7666667	0

➤ Moneyball - <http://www.youtube.com/watch?v=AiAHIZVgXjk>

➤ Moneyball (Breaking Biases) -
<http://www.youtube.com/watch?v=yGf6LNWY9AI>



Actividad 3.2 · Model Building + Role Play

Objetivo

Pasar de un problema real a una definición estructurada del modelo: datos → población → variables → target → implementación.

Se le pide que diga cuántos coches en movimiento hay en una calle de 2 sentidos de 1 km de largo en este momento. No tienes imágenes satelitales, lo único que tienes es una moto para recorrer la calle y contar los carros. No se puede dar un número exacto de autos porque todos los autos se están moviendo, por lo que su trabajo es dar su mejor estimación. Defina cómo recolectaría los datos (usando la motocicleta, puede viajar por la calle 0 veces, en 1 sentido o en ambos sentidos o muchas veces) y luego defina cada etapa del esquema del modelo a continuación y analice una o dos de sus principales limitaciones (para cada etapa).

ACTIVIDAD 3.2

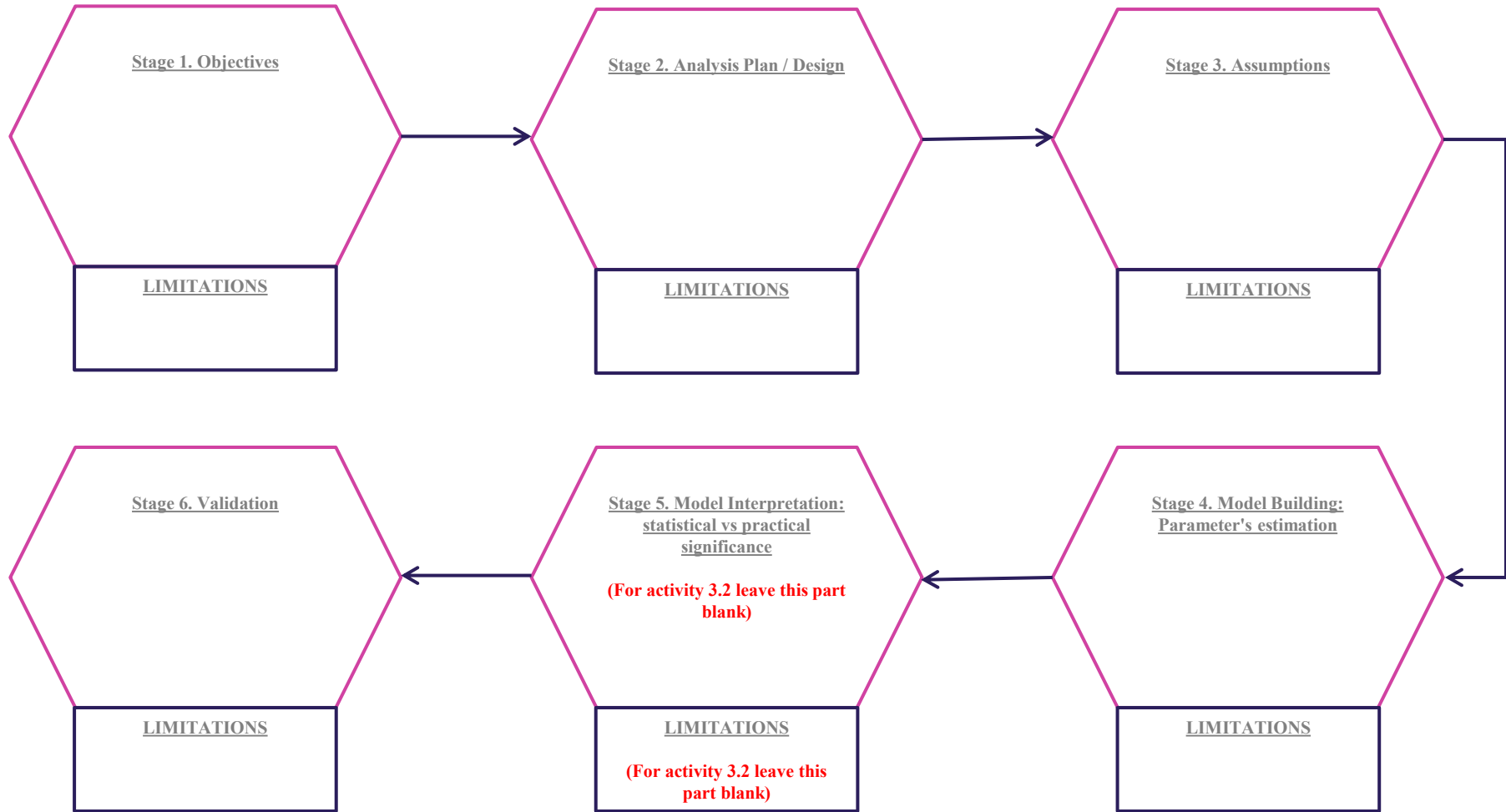
Pregunta guía

¿Cuántos coches en movimiento hay ahora mismo en una carretera de 1 km y dos sentidos?

No buscamos un número exacto: buscamos la mejor estimación posible y justificamos cómo recogeríamos los datos.

Entrega: PPTX o PDF usando el esquema proporcionado

ACTIVITY 3.2 – A SIX-STAGE APPROACH TO MULTIVARIABLE MODEL BUILDING



Actividad 3.2 · Ejercicio 1

¿Qué tienes que hacer?

- 1 Define cómo recogerías los datos con la motocicleta.
- 2 Puedes recorrer 0, 1, ambos sentidos o muchas veces.
- 3 Define cada etapa del esquema de Model Building.
- 4 Analiza 1–2 limitaciones principales de cada etapa.
- 5 Deja la etapa 5 en blanco para este ejercicio.

Piensa como un modelador

PROBLEMA → **DATOS** → **MODELO**

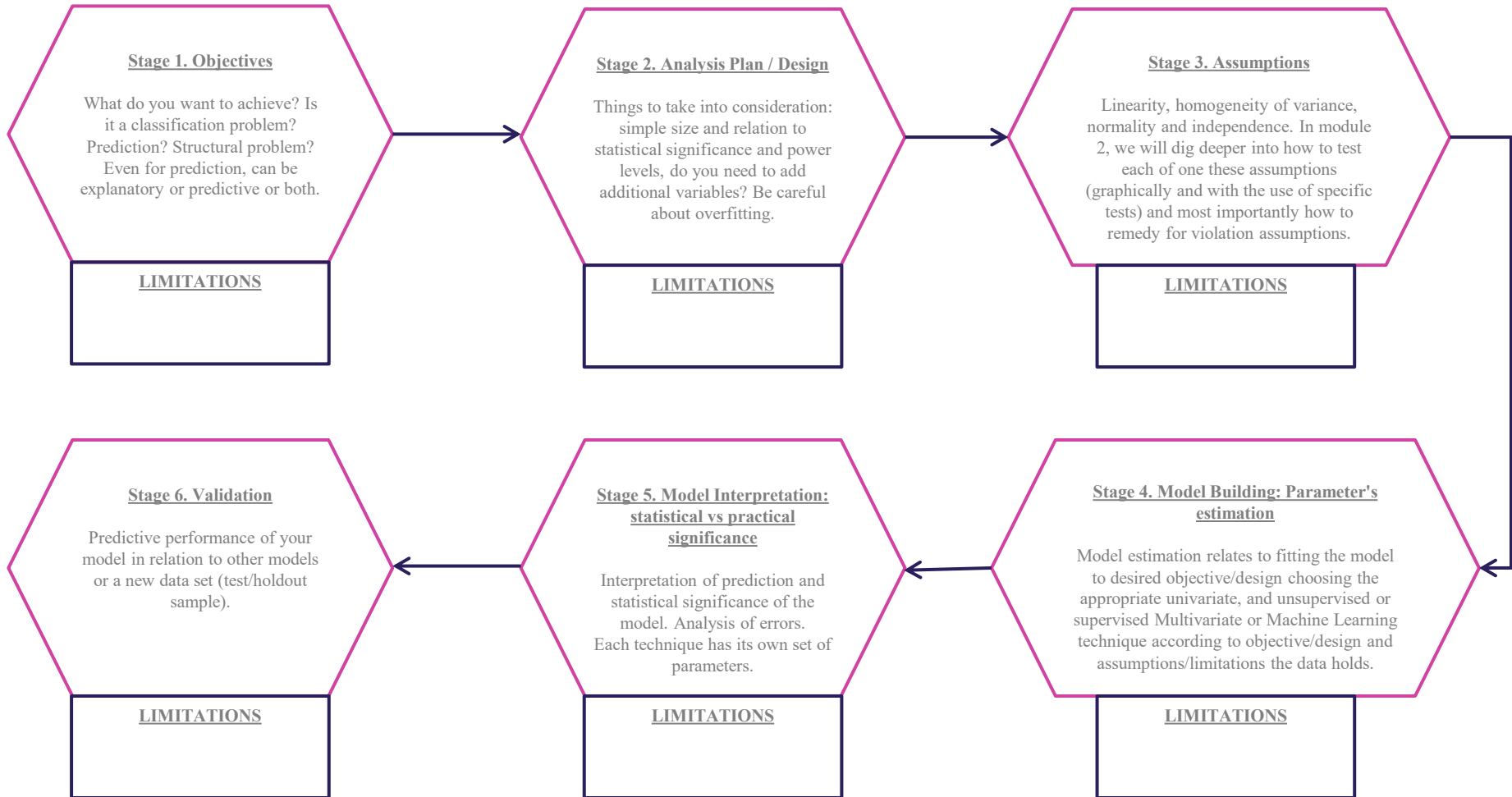
Tu respuesta debe justificar las decisiones, no solo enumerarlas.

Entrega

- PPTX o PDF
- Utiliza el esquema proporcionado

 **< 10 min**

SIX-STAGE APPROACH TO MULTIVARIABLE MODEL BUILDING

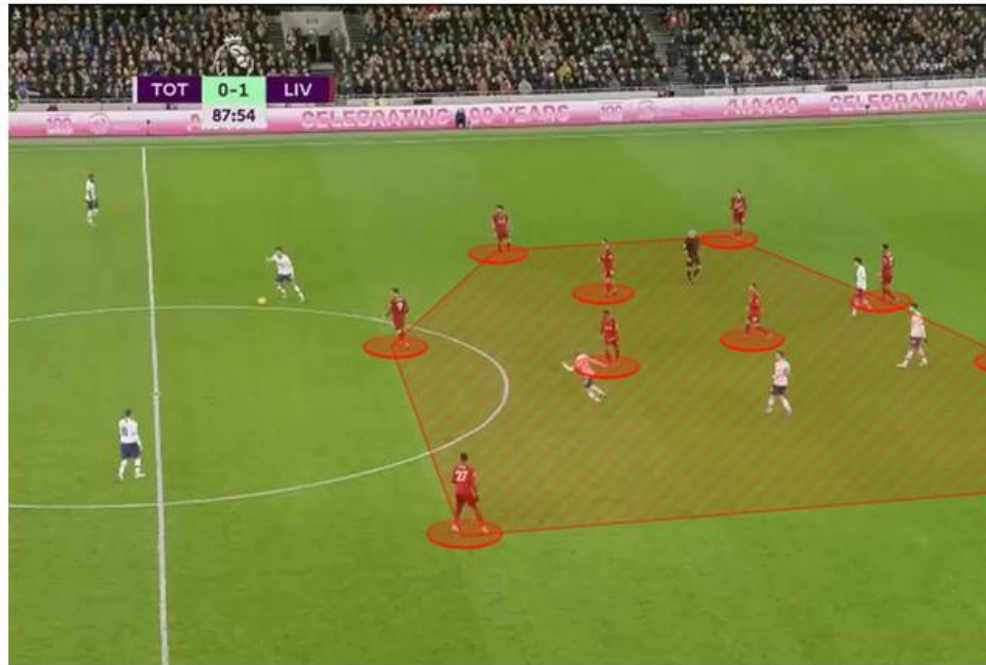


Football Example



Jurgen Klopp

Liverpool are using incredible data science during matches, and effects are extraordinary

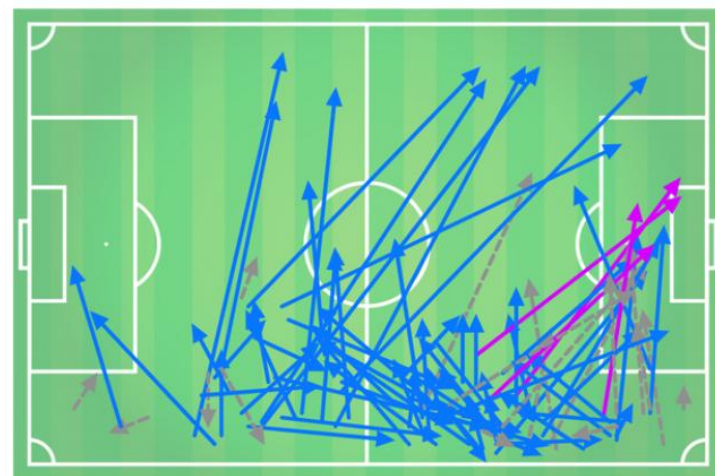
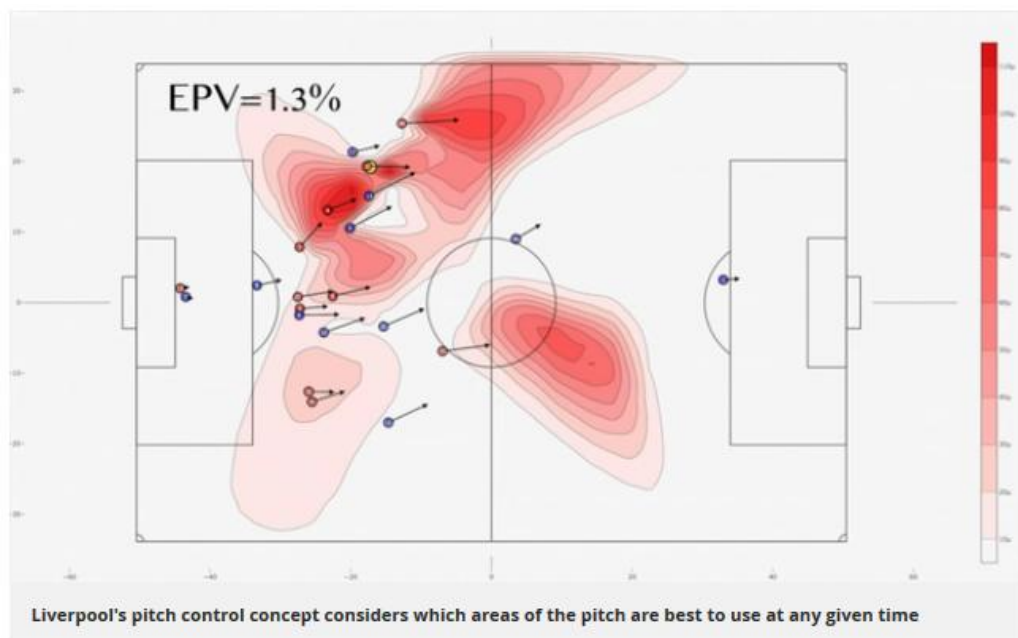


Liverpool form a compact block in the centre of the pitch late in the match with Spurs

source:

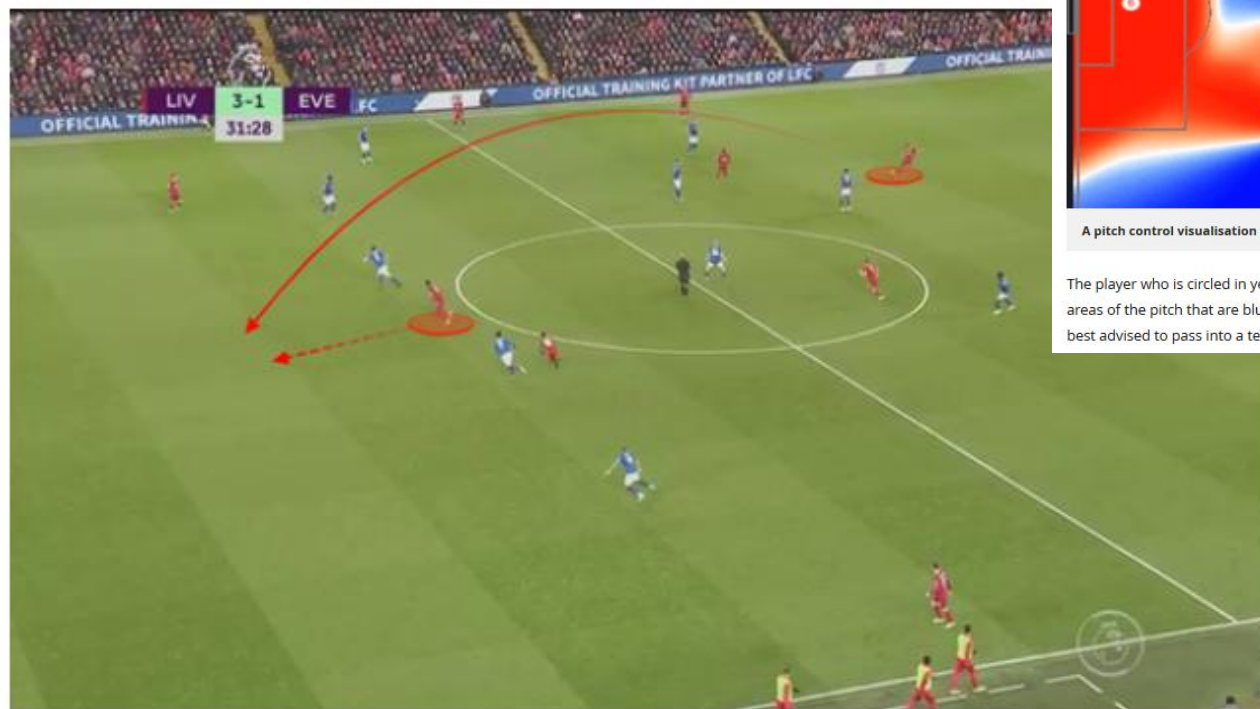
<https://www.liverpool.com/liverpool-fc-news/features/liverpool-transfer-news-jurgen-klopp-17569689>

Liverpool are using incredible data science during matches, and effects are extraordinary

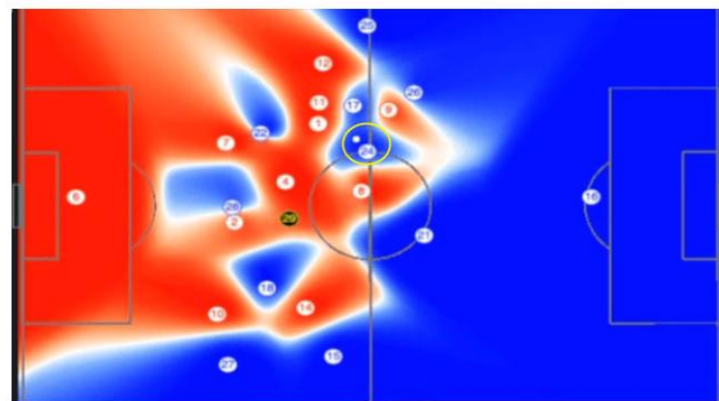


$$f(X) = y$$

Liverpool are using incredible data science during matches, and effects are extraordinary



Liverpool score against Everton by hitting a simple long pass over the defence



A pitch control visualisation which captures the regions of space controlled by certain players

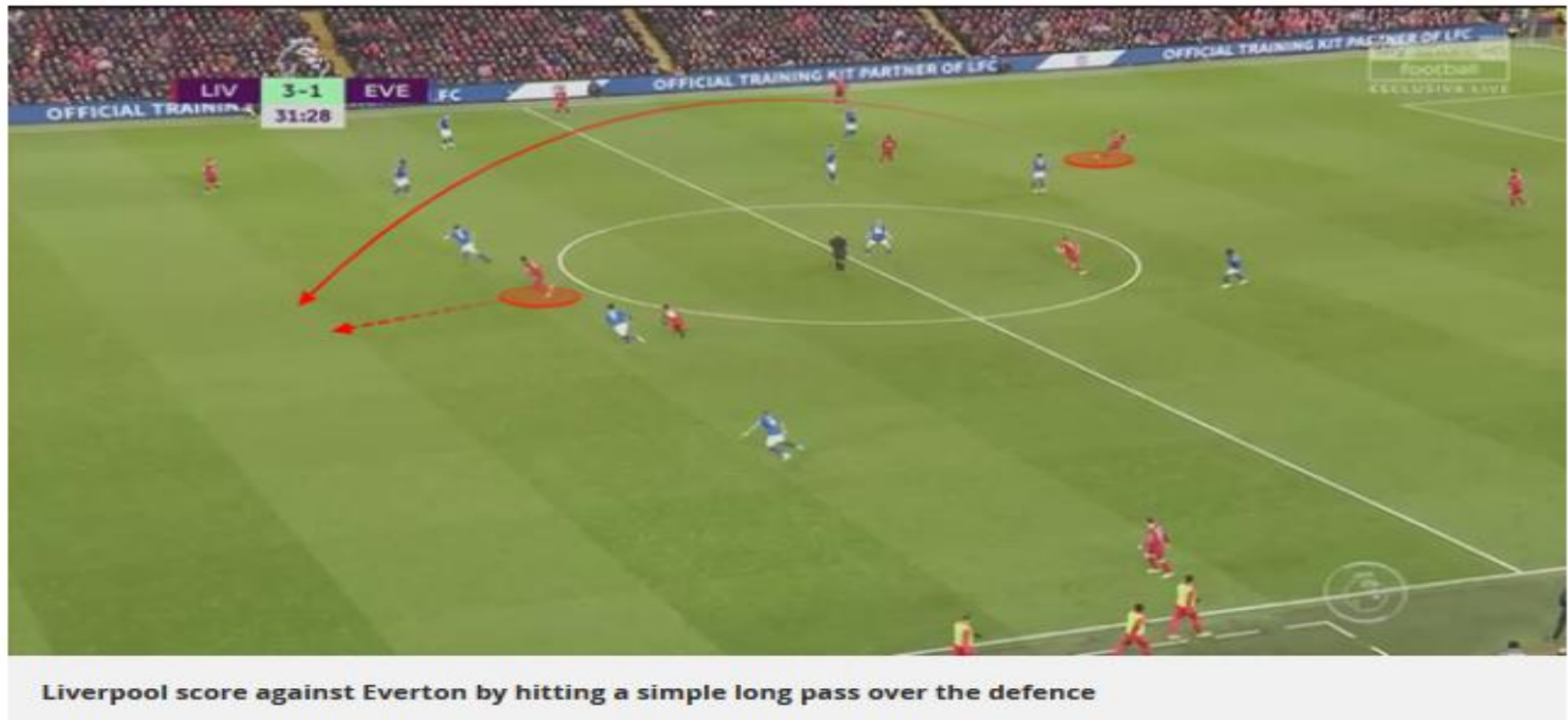
The player who is circled in yellow has possession of the ball and, essentially, his team has access to the areas of the pitch that are blue, while the red areas are mostly controlled by the opposition. The player is best advised to pass into a teammate positioned within a blue zone.



Jurgen
Klopp

Liverpool are Using Incredible Data Science

Actual footage:



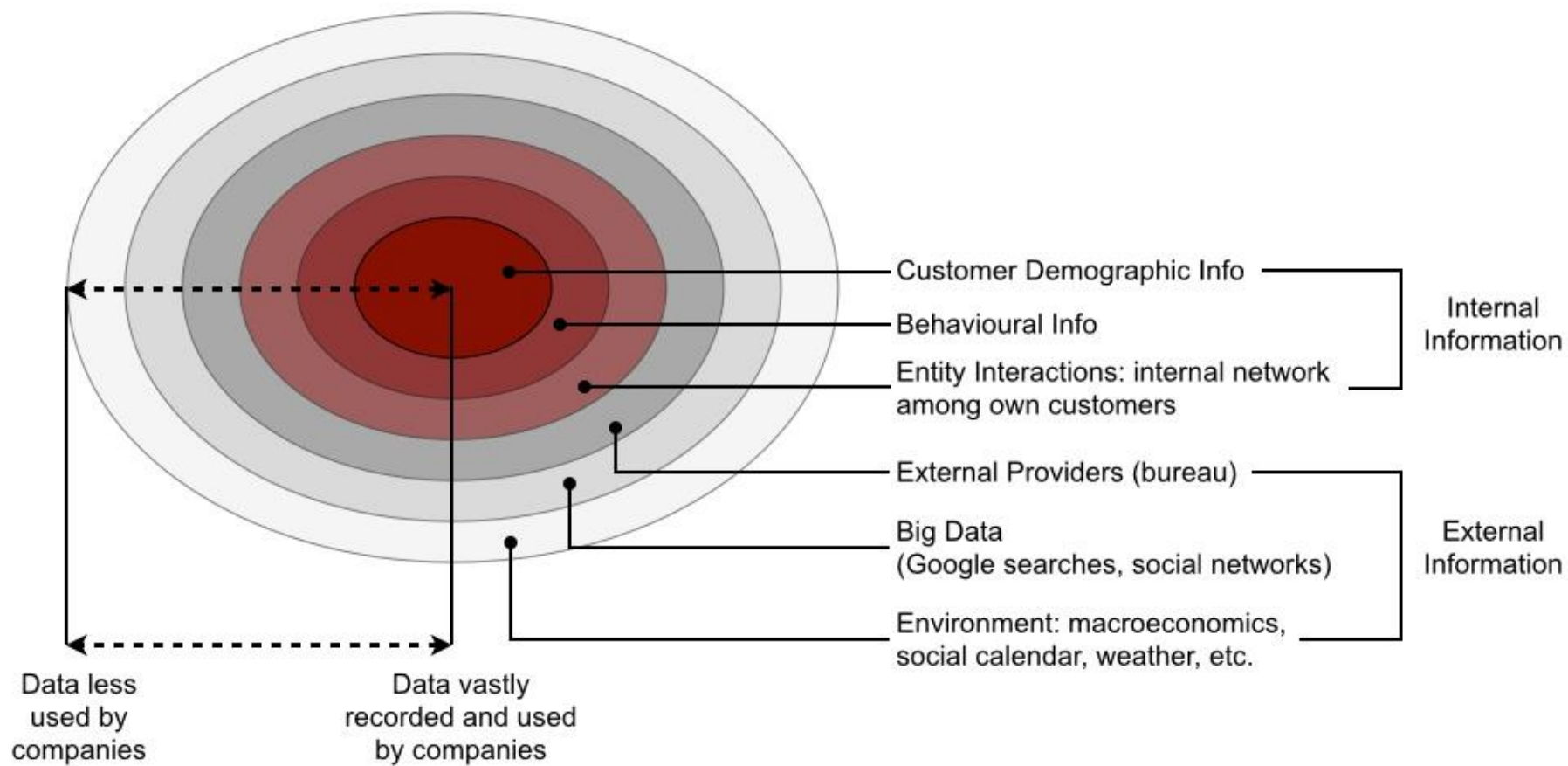
<https://www.youtube.com/watch?v=rYJD4MMfv1M&t=197s>

Integrated vision of the Portfolio Management approach



X

Map of information available for businesses



Machine Learning - Model estimation

$$f(X) = y$$

f (*X*) = *y*

ib_var_1	ib_var_2	ib_var_3	icn_var_22	icn_var_23	icn_var_24	ico_var_25	ico_var_26	ico_var_27	if_var_65	if_var_68	if_var_80	ob_target
0	1	0	2	1	2	4	4	4	31	4626	0.7666667	0
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0	0	0	3	3	1	3	4	5	55	3084	0.6	1
0	1	0	2	1	1	4	5	4	45	616.8	0.7666667	0

Machine Learning - Model estimation

$$f(X) = y$$

X = data frame with observed characteristics and data points.

y = target variable, variable we wish to predict

ROLE PLAY · Round 1

Busca una empresa que aplique Machine Learning a un problema de negocio.

NO REPITAS

Caso, vídeo, compañeros anteriores o el mismo problema.

CUALQUIER PAÍS

Puedes repetir país y sector.

≤ 280 CARACTERES

Sé muy breve y concreto.

Formato de respuesta

País · Sector · Empresa · Problema que intenta predecir o resolver

Ejemplo: España · Banca · Banco Santander · Recobro y recuperación

ROLE PLAY – Rounds 1, 2 and 3 - Round 1

Country: Spain

Company: Banco Santander

Task: Collection and Recovery activities

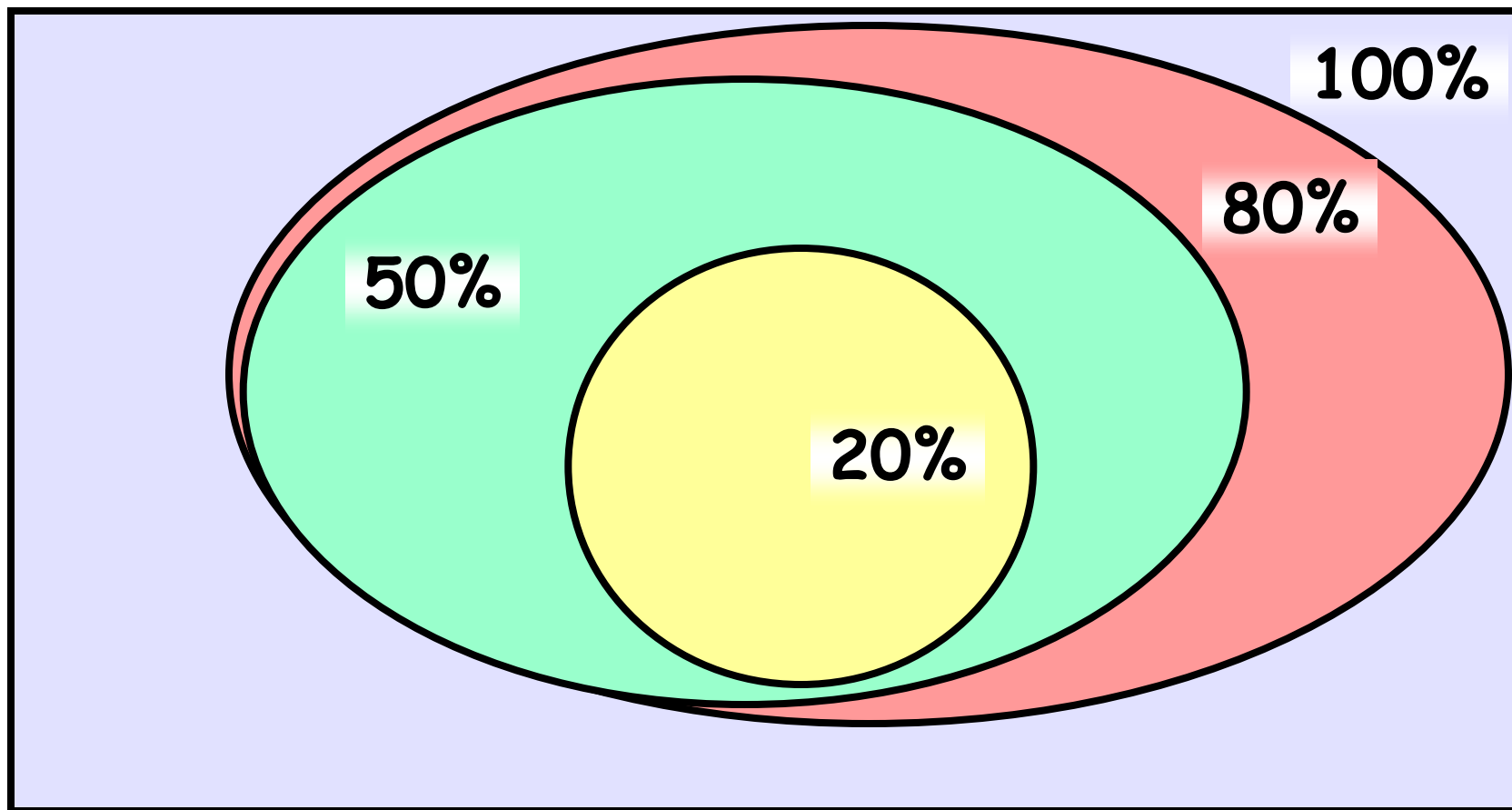
Machine Learning: supervised – classification

Every Human

Living in Spain

Existing Customers

Default (Unpaid Debt)



Population: Existing Customers in Spain with Unpaid Debts.

ROLE PLAY · Round 2

ROUND 2

Completa la definición del modelo

X · Población	¿A quién / qué se aplicará el modelo?
X · 3 features	Tres variables de entrada plausibles.
y · Target	Define exactamente qué significa 1 y 0.

No dejes la respuesta en blanco: puedes inventar una definición razonable si no conoces el modelo real.

ROLE PLAY – Rounds 1, 2 and 3 - Round 2

X:

- **Population:**
- **Characteristics:**

y:

- **Target definition:**

ROLE PLAY – Rounds 1, 2 and 3 - Round 2

$$f(X) = y$$

X:

- **Population:**

Existing Customers in Spain with Unpaid Debts

- **Characteristics:**

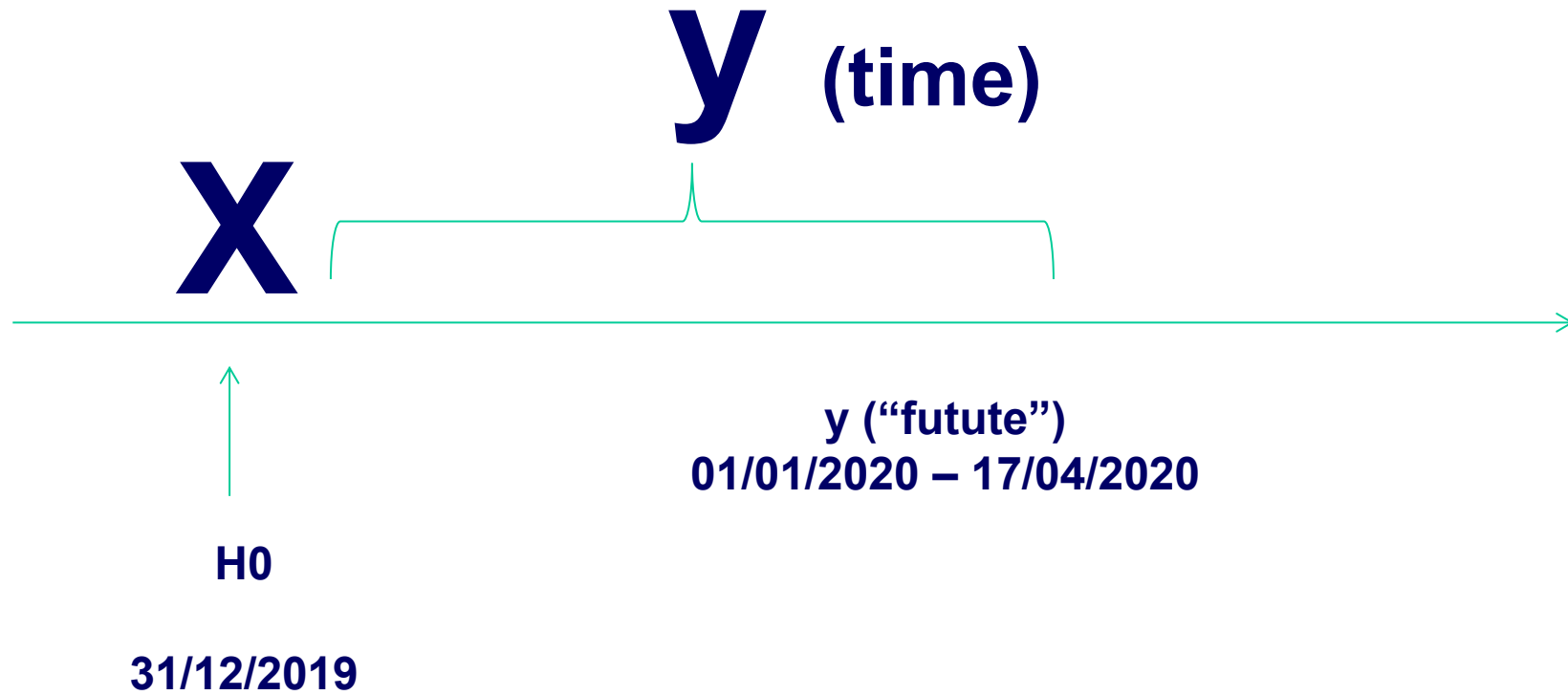
- Time on books
- Total Paid last 12 months
- Total Debt in Arrears

y:

- **Target definition:**

1 – good – customer who repaid more than 90% of the debt in arrears the following 6 months counting from the payment due date.

0 – bad – every one else.



Machine Learning - Model estimation

$$f(X) = y$$

Characteristics (select)

Target
definition
(otro select)

f (*X*) = *y*

Population (where)

ib_var_1	ib_var_2	ib_var_3	icn_var_22	icn_var_23	icn_var_24	ico_var_25	ico_var_26	ico_var_27	if_var_65	if_var_68	if_var_80	ob_target
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0	1	0	2	1	1	4	5	4	45	616.8	0.7666667	0

35

ROLE PLAY · Round 3

PROMPT:

Create a python code for a logistic regression using four columns data frame.

First column discrete age, second column salary as a continuous variable and third column gender binary variable. Forth column, the target variable should be binary pay or not pay the loan.

Generate all variables randomly using numpy. Set the seed so that all variables always get the same value. Save the data to an excel named

unav_pwc.xlsx without index. Using joblib, dump the logistic regression model as **unav_pwc.joblib**

Show the professor the 3 elements;

- the python code (.ipynb)
- the excel file generated by the code (.xlsx)
- the model dumped by the code (.joblib)

ROUND 3

Genera datos sintéticos + modelo

Pide a ChatGPT (o similar) que genere:

- código Python de una regresión logística
- edad · salario · género · target binario
- variables aleatorias con NumPy + seed
- Excel: psdma.xlsx sin índice
- modelo Joblib: psdma.joblib

ENTREGA · 4 elementos

.ipynb · .xlsx · .joblib · tu participación

ROLE PLAY – Rounds 1, 2 and 3 - Round 2

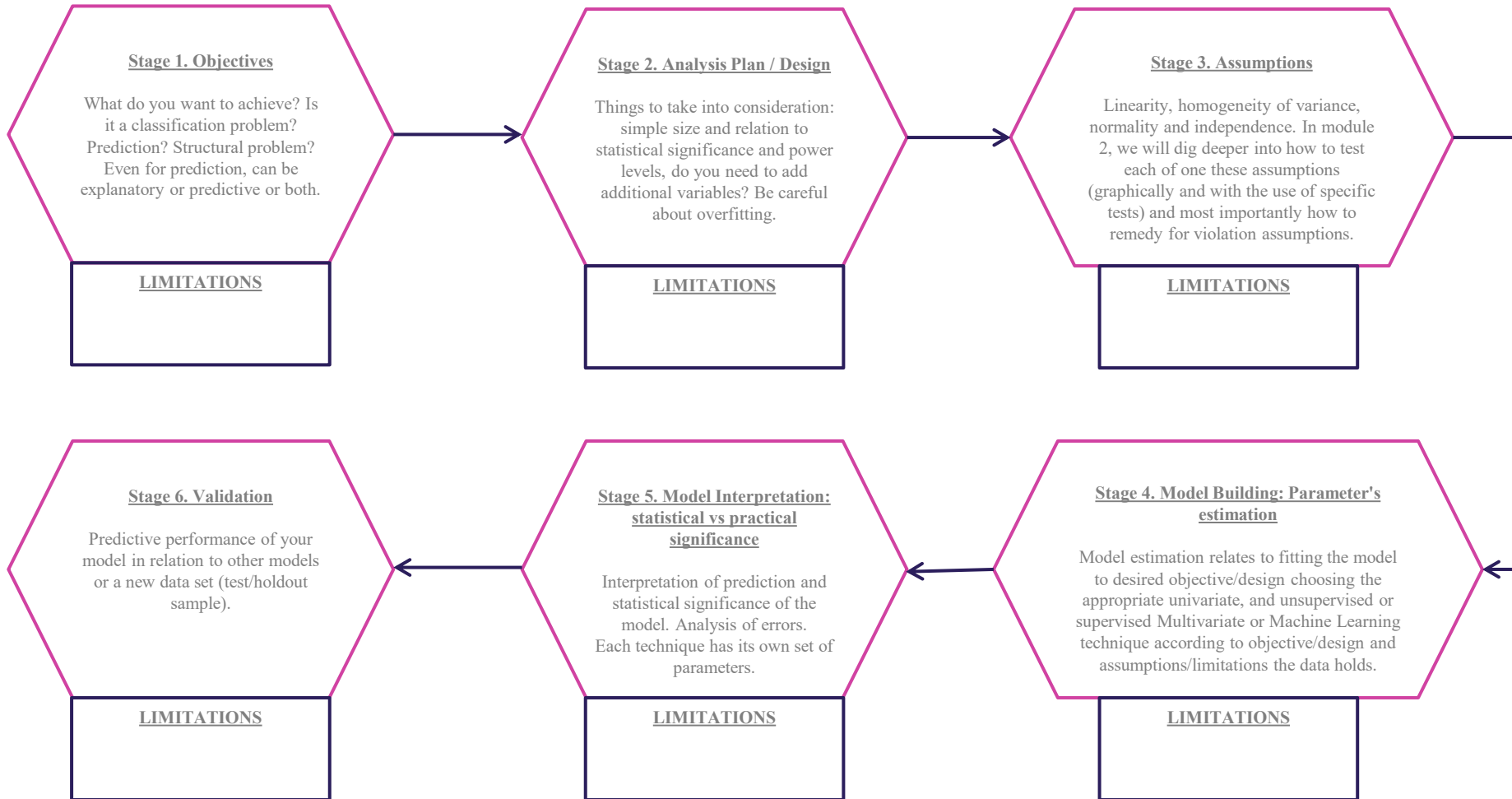
$$f(X) = y$$

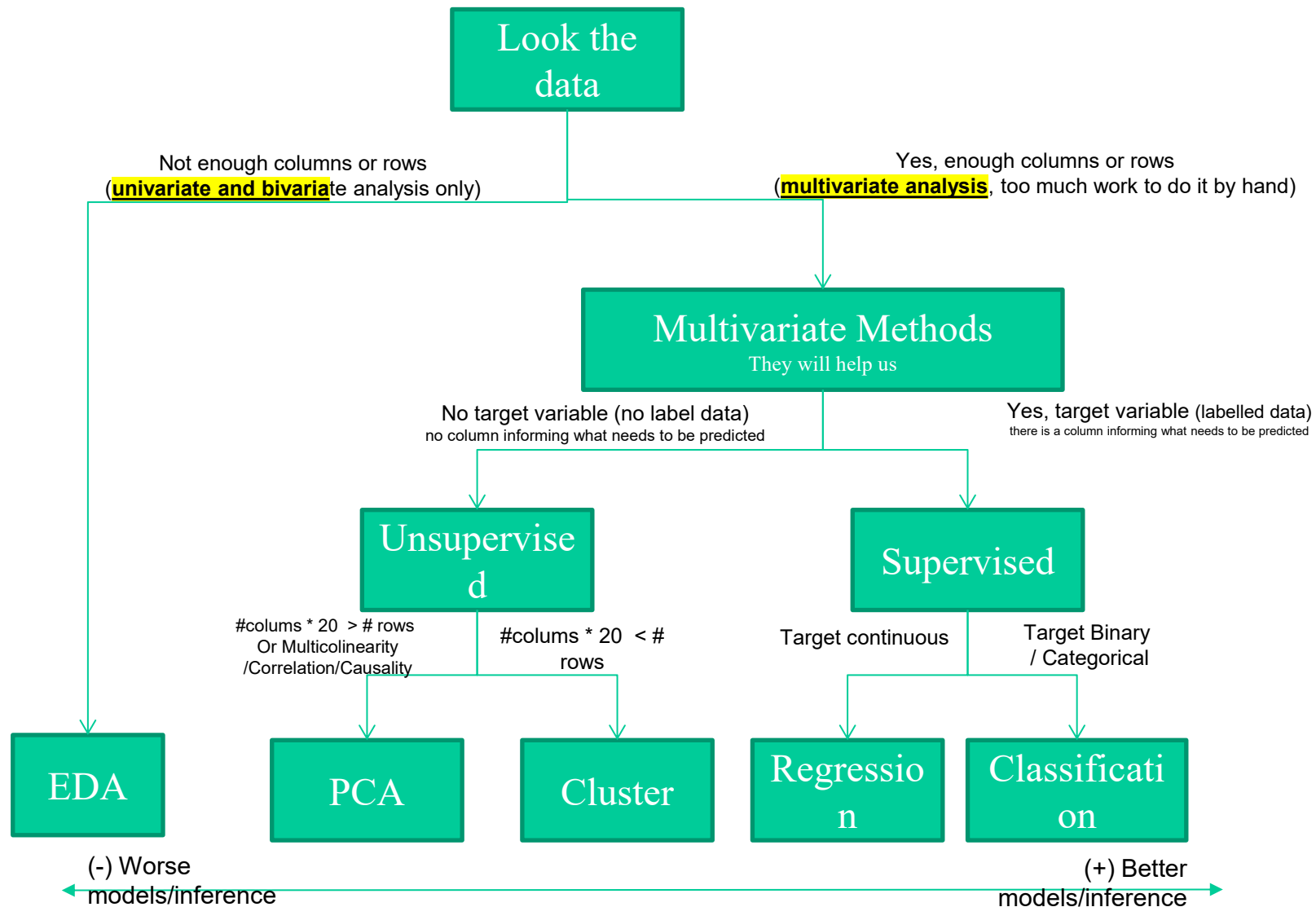
$$X \& y = .xlsx$$

$$f = .ipynb$$

$$f^* = .joblib \text{ (modelo entrenado)}$$

SIX-STAGE APPROACH TO MULTIVARIABLE MODEL BUILDING







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PSDMA - MOTIVATION

Why the PSDMA is fundamental part of the degree? PSDMA is not a typical statistical course because it is agnostic:

- Statistical and Machine Learning cultures are typically different.
- **Statistical** approach focus in discovering the underlying structure of the problem, like in physics finding the formula to calculate the speed of light based on the material its going through, here the key is the validity of the formula, not the calculated speed of light.
- **Machine Learning** approach focus on the output of the formula, in other words the calculated value or the prediction of the model, in our physics example it would mean focusing on the correct value of the speed of light not in the validity of the formula involved in the calculation.

SPITFIRE EXAMPLE

Let's suppose during the World War II, a general want to analyse our spitfire fighting and decide which parts of the airplane it should reinforce



SPITFIRE EXAMPLE

Statistical Approach

A typical statistician design the following controlled example:

- Send several spitfire planes to fight and evaluate where they got hit and whether they shot down or not.



Discussion:

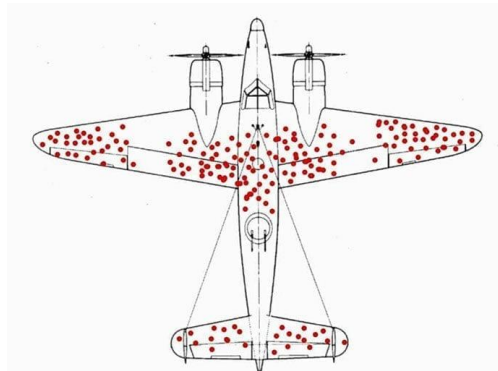
Is this approach correct? Is this approach feasible? How do you rescue planes?

Multivariate Analysis

SPITFIRE EXAMPLE

Real Case

During World War II, the Allies studied Nazi damage to their airplanes. Their study resulted in this dotted illustration:



The analysis of these damages resulted in the idea that their airplanes should be reinforced in the areas of the airplane showing the densest clusters of red dots.

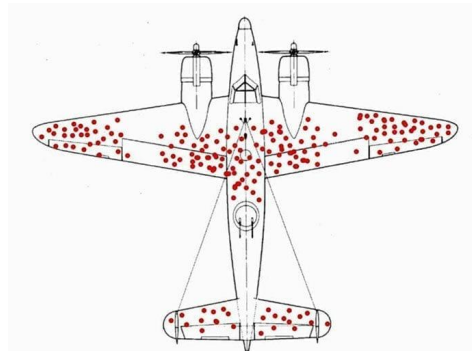
Discussion: is this approach

correct?

SPITFIRE EXAMPLE

Real Case

During World War II, the Allies studied Nazi damage to their airplanes. Their study resulted in this dotted illustration:



A mathematician by the name Abraham Wald didn't agree. Instead, he suggested that the airplanes should be reinforced where there were no indications of damage.

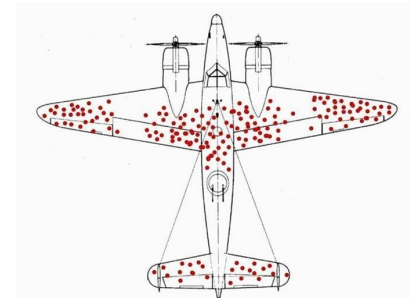


<https://www.youtube.com/watch?v=ZyLVlvBidIA>

Multivariate Analysis

SPITFIRE EXAMPLE

Real Case - solution



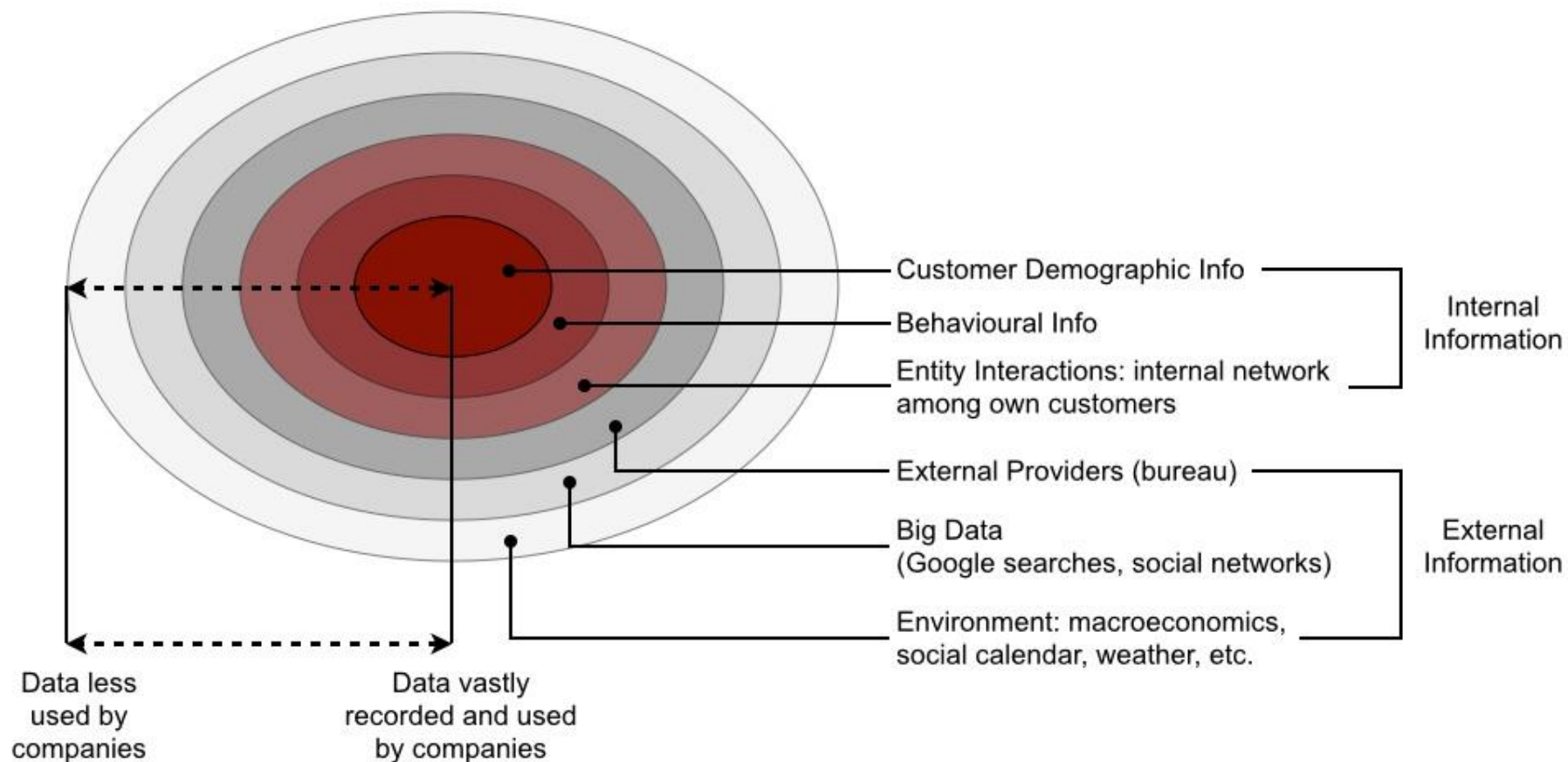
Wald was right, of course. By studying planes that got back home somehow, the red areas indicated non-fatal damage areas. If it had been possible to study the plane that were shot down and destroyed by the Nazis, they would be likely to have found an inverse pattern.

The fallacy of survivorship bias

This is a great example of a very specific fallacy — [survivorship bias](#).

EXAMPLES OF DATA COMPANIES USE

Map of information available for businesses





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Artificial General Intelligence (AGI)

AGI is a machine with general intelligence much like a human being to solve any problem as humans do in any given situation.

AGI characteristics (those we see in all humans):

- Common Sense
- Background Knowledge
- Transfer Learning
- Abstraction
- Causality



Point of view A: a man apparently attacking another man



Point of view B: a man saving another man from an accident

AI researchers and scientists need to find a way to make machines conscious, programming a full set of cognitive abilities.

- Fujitsu-built K, built one of the fastest supercomputers also a notable attempt at achieving AGI, system took 40 minutes to simulate a single second of neural activity!!



Natural language processing, understanding and generation

NLG

- Discourse Generation
- Lexical Choice
- Sentence Planning/ Generation
- Realization
- Document Structuring

NLP

- Named entity Recognition (NER)
- Part-of-Speech Tagging (POS)
- Syntactic Parsing
- Coreference Resolution
- Machine Translation

NLU

- Lexical Ambiguity/ Analysis
- Sentiment Analysis
- Topic Classification
- Information Extraction
- Entity Detection
- Summarization
- Semantic Parsing
- Syntactical Analysis

- Has ChatGPT Achieved Artificial General Intelligence? -

<https://www.youtube.com/watch?v=JpmUYoN8NKY>

Low Code Exercise

Divide the class in several groups

Open ChatGPT and post the following prompt

Create a python code for a statsmodels logistic regression using the variables Pclass, Sex and Embarked from the titanic dataset directly from an internet source or github, transform the variables in dummies. Set the seed so that all variables always get the same value. Split the data into train and test. Print the model summary.

Low Code Exercise

Copy the code into Google Colab

<https://colab.research.google.com/>

```
import pandas as pd
import statsmodels.api as sm
from sklearn.model_selection import train_test_split

# Set seed for reproducibility
seed = 0

# Load the Titanic dataset from an internet source or github
url = 'https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv'
df = pd.read_csv(url)

# Create dummy variables for categorical variables
df = pd.get_dummies(df, columns=['Pclass', 'Sex', 'Embarked'], drop_first=True)

# Split the data into train and test sets
X = df[['Pclass_2', 'Pclass_3', 'Sex_male', 'Embarked_Q', 'Embarked_S']]
y = df['Survived']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=seed)

# Fit a logistic regression model using statsmodels
model = sm.Logit(y_train, sm.add_constant(X_train)).fit()
print(model.summary())
```

Low Code Exercise

Divide the class in several groups

Optimization terminated successfully
Current function value:
Iterations 6

Logit

=====

Dep. Variable: Surv

Model: L

Method:

Date: Thu, 29 Jun

Time: 16:2

converged:

Covariance Type:

nonrobust LLR p-value: 3.252e-58

=====

	coef	std err	z	P> z	[0.025	0.975]
const	2.4459	0.294	8.313	0.000	1.869	3.023
Pclass_2	-0.6058	0.279	-2.172	0.030	-1.152	-0.059
Pclass_3	-1.7074	0.247	-6.925	0.000	-2.191	-1.224
Sex_male	-2.5618	0.204	-12.566	0.000	-2.961	-2.162
Embarked_Q	0.0504	0.395	0.127	0.899	-0.724	0.825
Embarked_S	-0.4610	0.254	-1.818	0.069	-0.958	0.036

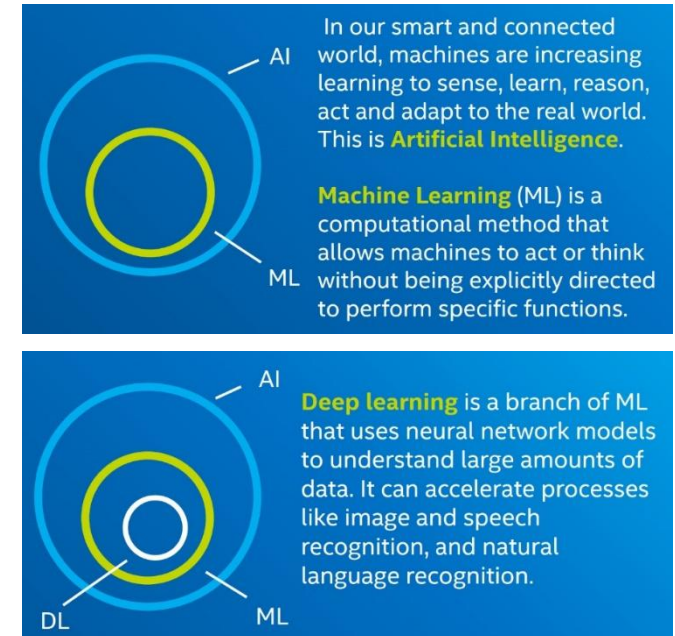
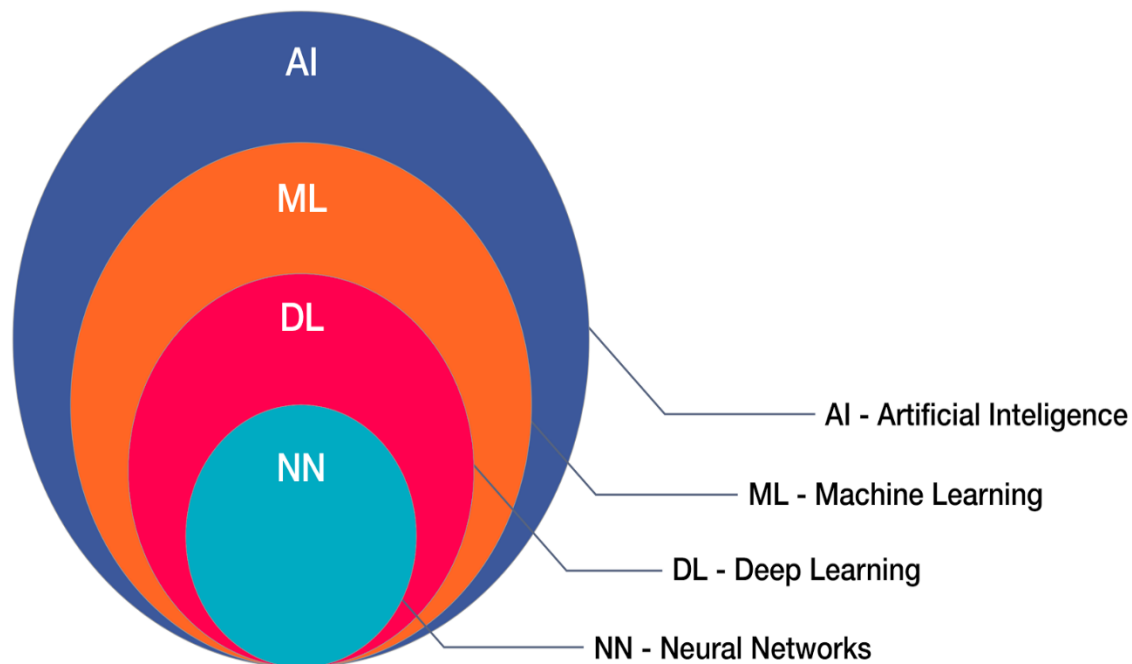
=====

	Rose	Jack	Fiancé
			
Title	Miss.	Mr.	Mr.
Sex	1 (female)	0 (male)	0 (male)
Age	17	20	24
Fare	280 (high fare)	7.8 (low fare)	280 (high fare)
Family Size	4 (mother, fiancé and fiancé's guard)	2 (one friend)	4 (Rose, Rose's mother, his guard)
Embarked	Southampton	Southampton	Southampton
Special title	0 (Just Miss.)	0 (Just Mr.)	0 (Just Mr.)



Universidad de Navarra

AI taxonomy



AI types based on capabilities:

- Artificial narrow intelligence (ANI)- Narrow range of abilities.
- Artificial general intelligence (AGI)-On par with human capabilities.
- Artificial superintelligence (ASI)-More capable than a human.

Artificial Intelligence eras



Software
Intelligence



Narrow
Intelligence



General
Intelligence



Super
Intelligence



Cognitive
Software
Applications



Simple
Neural
Network
Machines



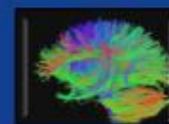
Deep
Learning
Machines



Deep
Reinforcement
Machines



Brain Inspired
Machines



Brain Sized
Machines



Super
Intelligent
Machines



Quantum
Intelligent
Machines

Software era
2012-2014

Learning era

2016

2018-2021

Cognitive era

2029-2040
Singularity

Quantum era

Computer vision: suspicious behavior in real time for retail

NTT East Japan & Earth Eyes camera using open source “AI Guardsman” from Carnegie Mellon



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Source: Futures Platform, July 2018

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Able to load all 350 passengers on an A380 within 20 minutes, 2x faster

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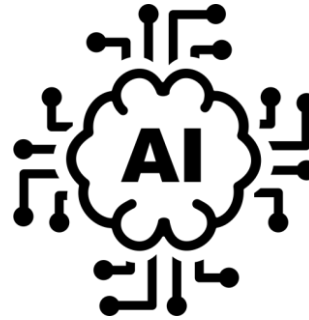


Source: Lufthansa, March 2018

What is AI?

Human approach:

- Systems that think like humans
- Systems that act like humans
- Systems that learn by themselves
- Systems that solve complex problems



Ideal approach:

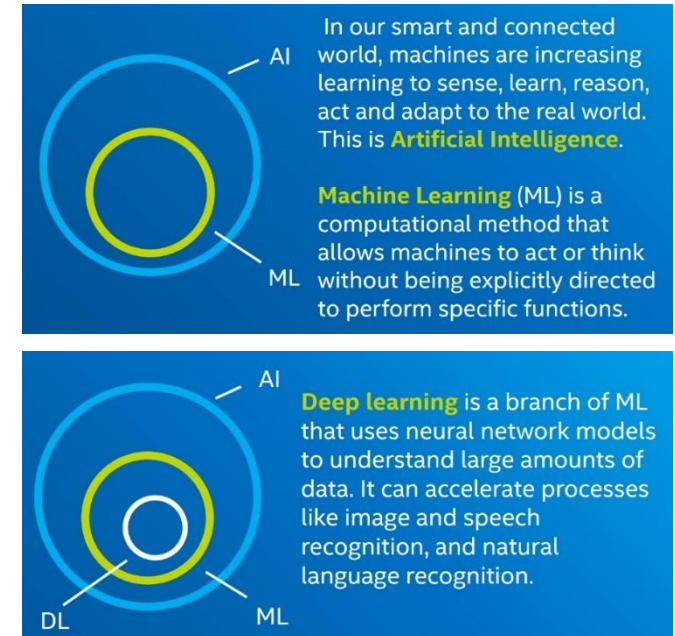
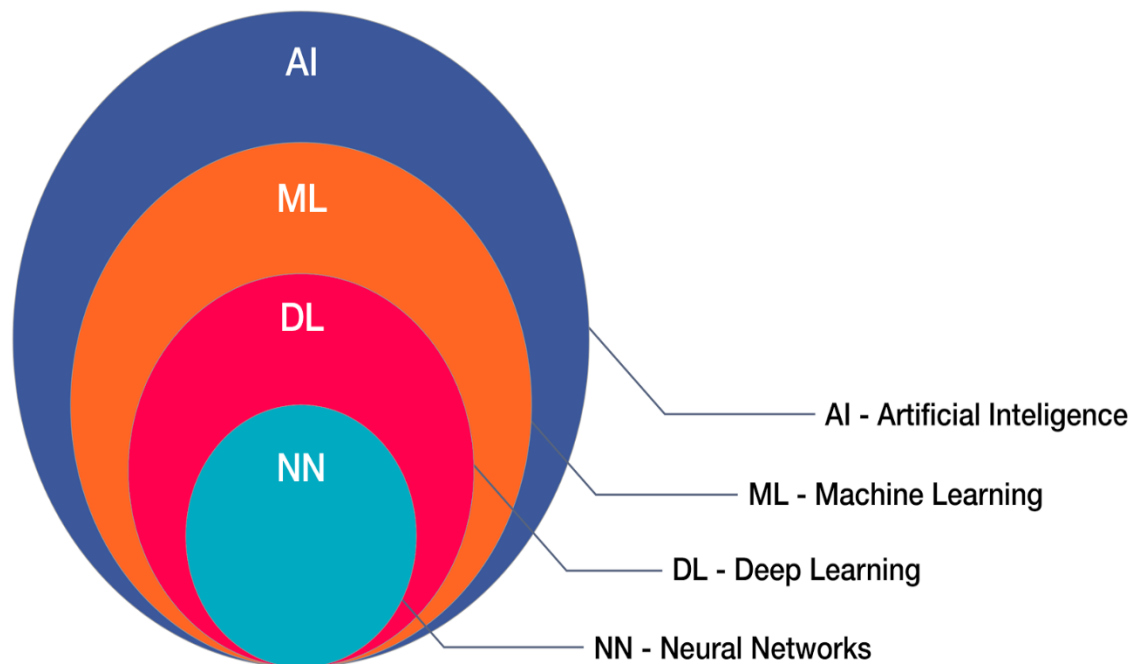
- Systems that think rationally
- Systems that act rationally

source: Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig

It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable.

source: Artificial Intelligence: What is artificial intelligence ?. John McCarthy (2004)

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Software
Intelligence



Narrow
Intelligence



General
Intelligence



Super
Intelligence



Cognitive
Software
Applications



Simple
Neural
Network
Machines



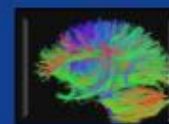
Deep
Learning
Machines



Deep
Reinforcement
Machines



Brain Inspired
Machines



Brain Sized
Machines



Super
Intelligent
Machines



Quantum
Intelligent
Machines

Software era
2012-2014

Learning era

2016

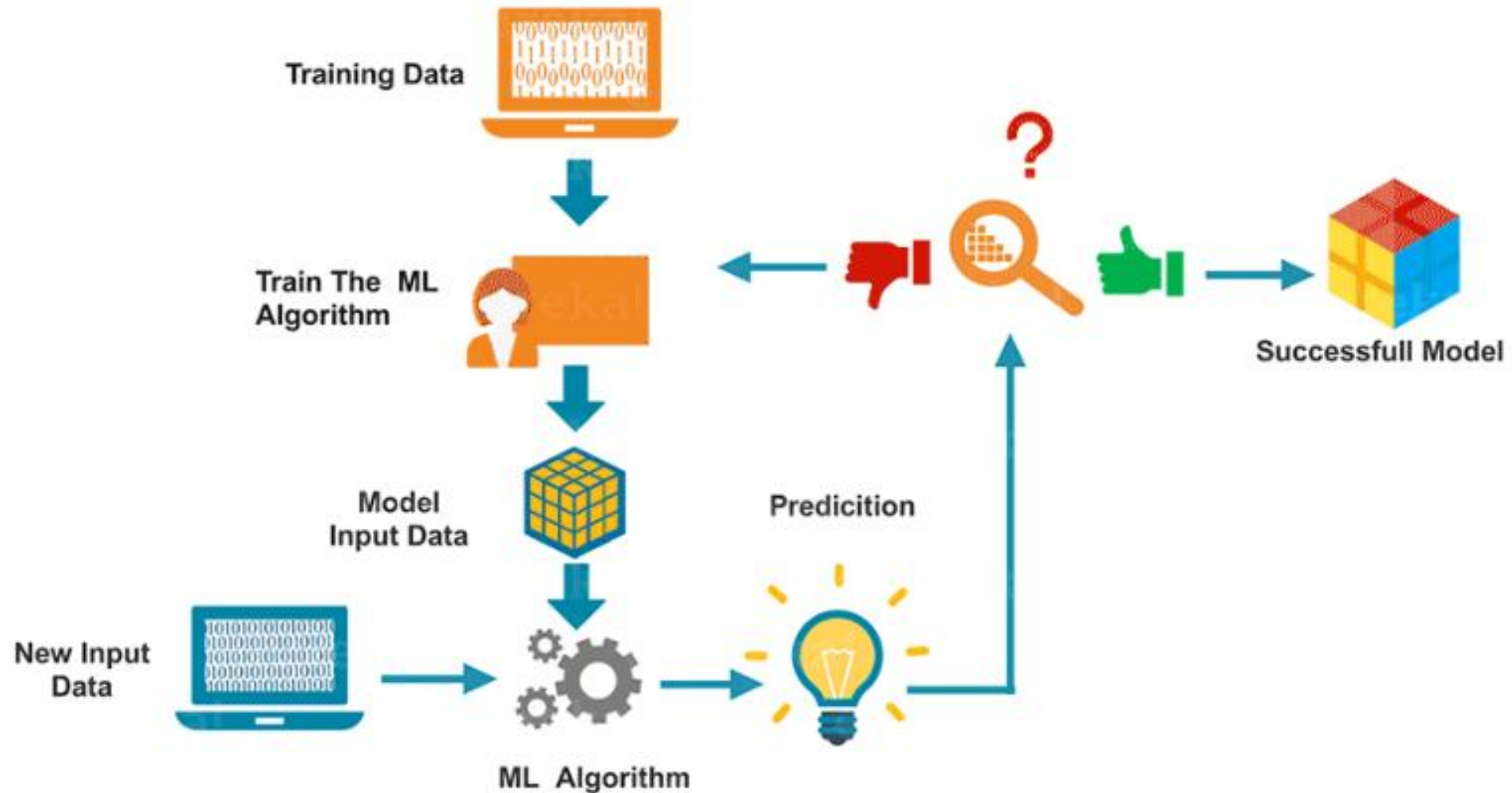
2018-2021

Cognitive era

2029-2040
Singularity

Quantum era

How does a machine learn work?



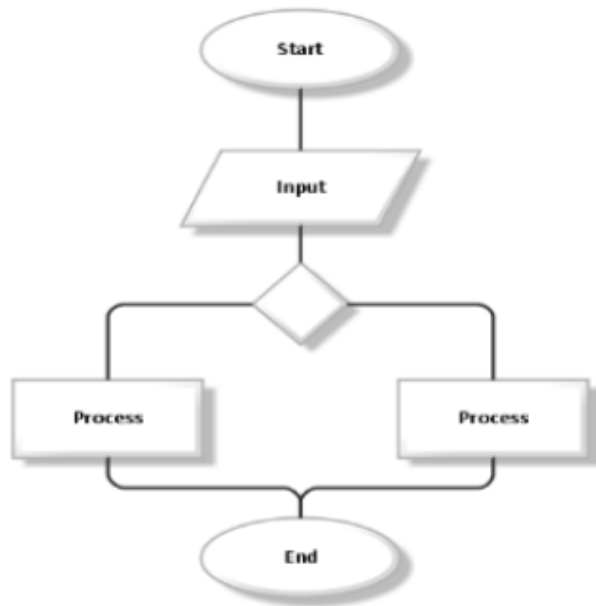
Machine learning automates the task of consuming and “understanding” data by training and some level of ‘thinking’

Machine learning is all about patterns recognition

Rules-based

“If this, then that”

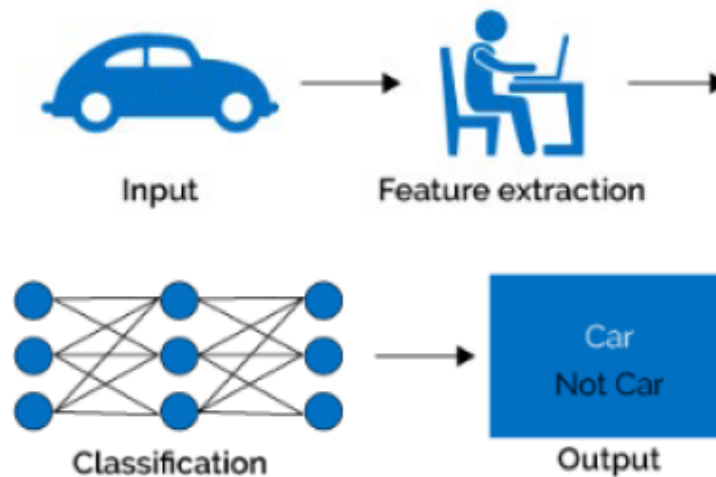
Limited by human imagination



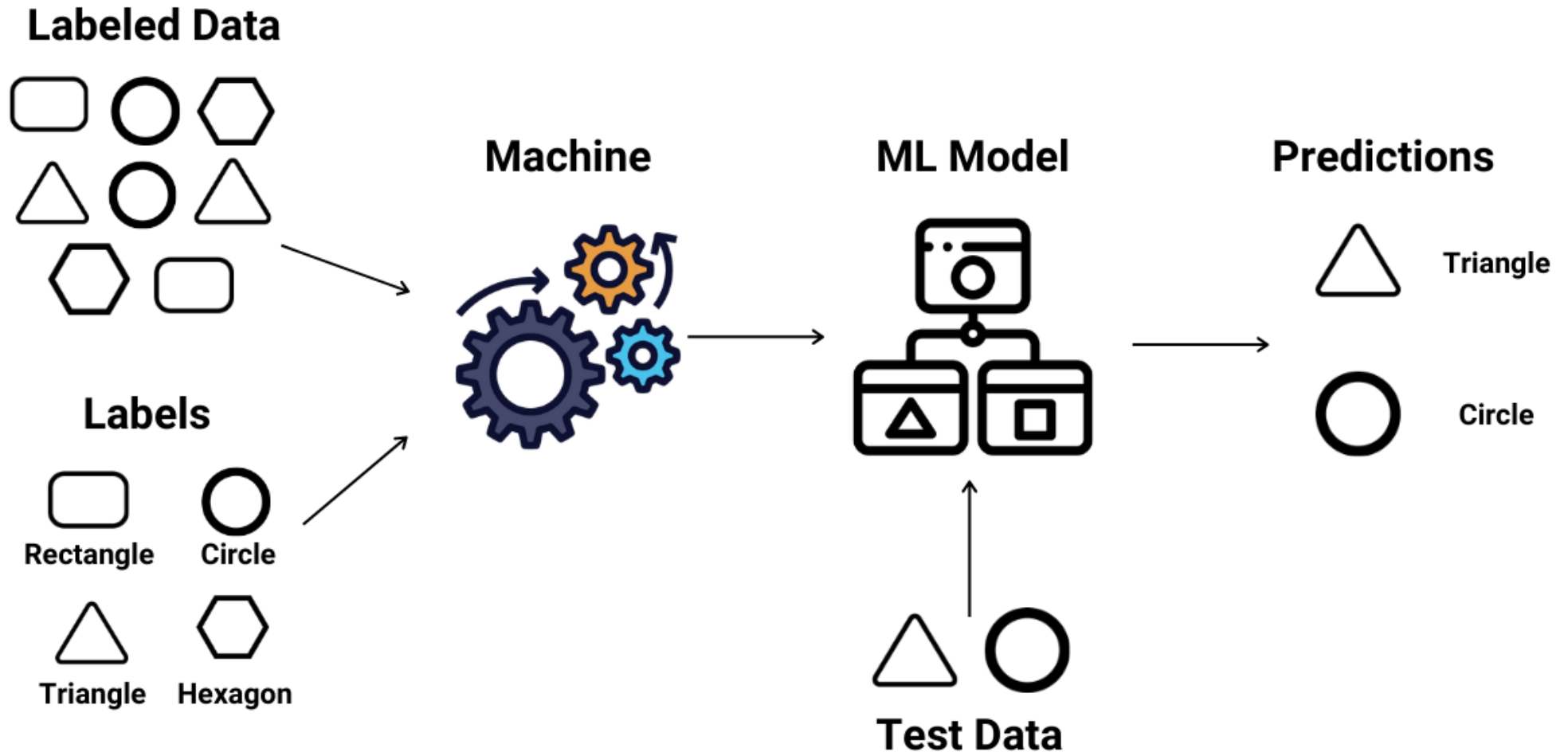
Algorithm-based, from iterative data analysis

The system figures out the rules

Far surpasses human limitations

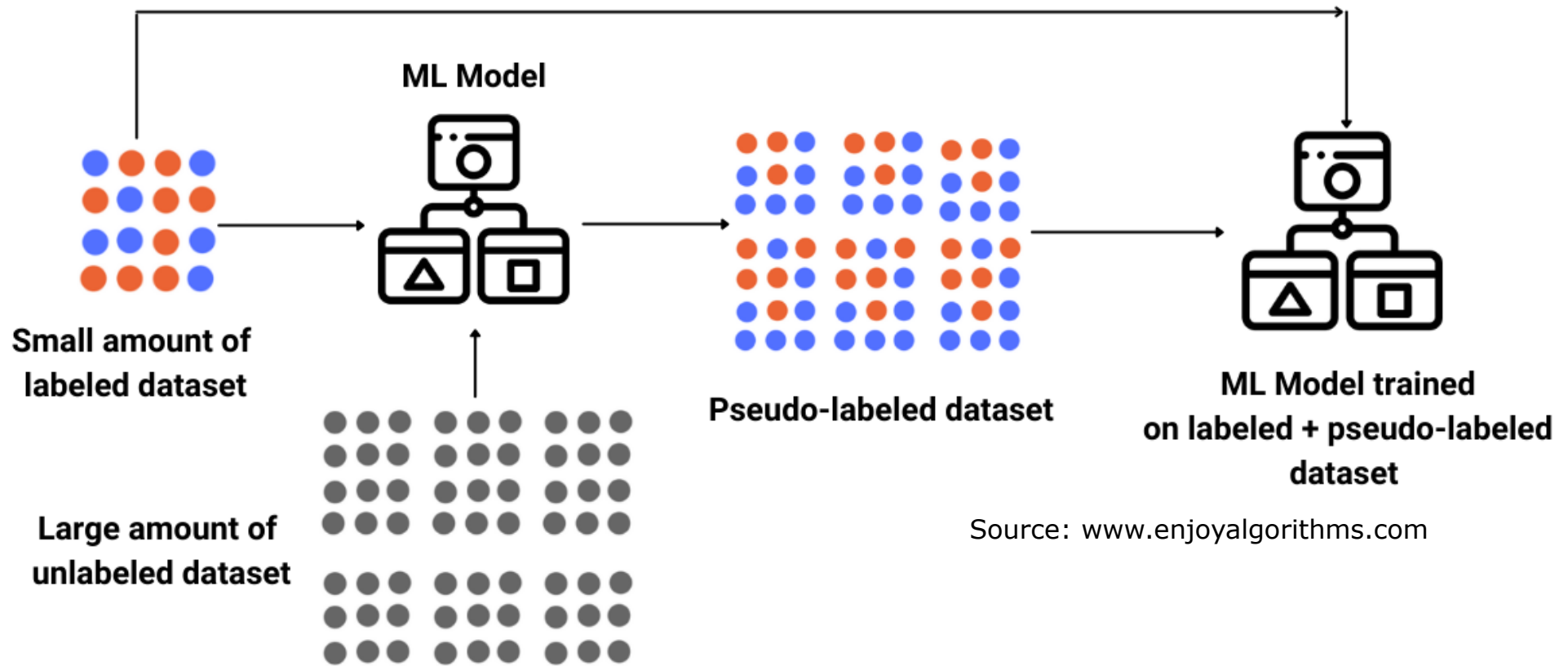


Machine learning. Supervised Learning



Source: www.enjoyalgorithms.com

Semi-Supervised machine learning

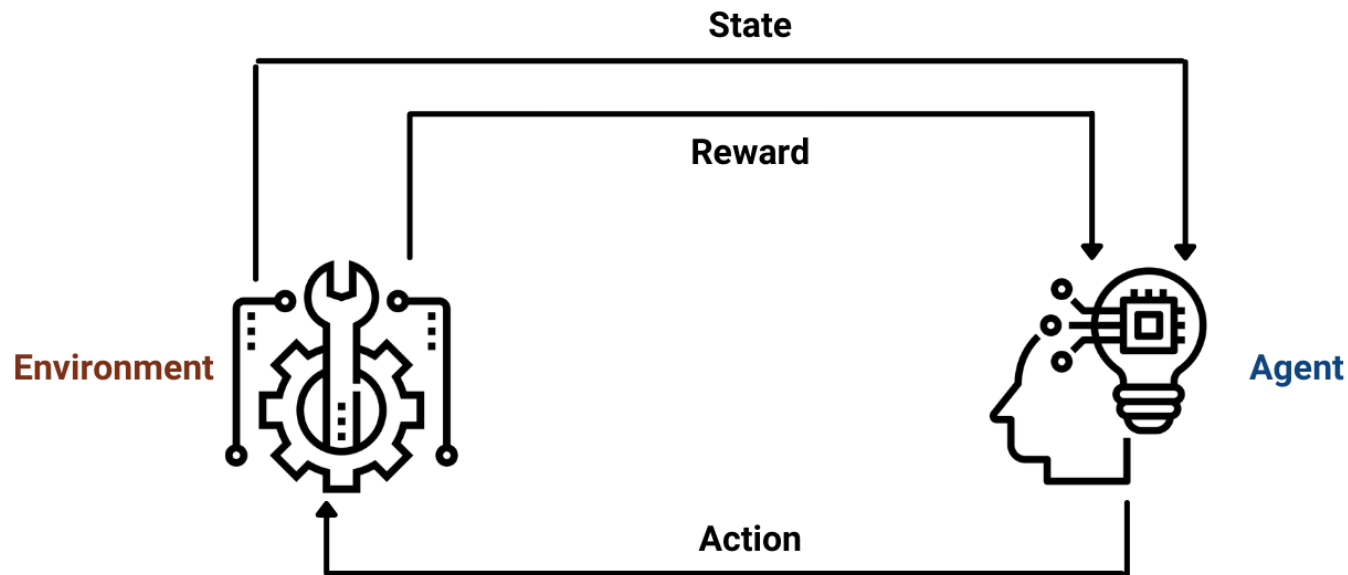


Semi-supervised learning is, for the most part, just what it sounds like: a training dataset with both labeled and unlabeled data. This method is particularly useful when extracting relevant features from the data is difficult, and labeling examples is a time-intensive task for experts.

Reinforcement Learning

Reinforcement learning is the training of machine learning models to make a sequence of decisions. The agent learns to achieve a goal in an uncertain, potentially complex environment. In reinforcement learning, an artificial intelligence faces a game-like situation. The computer employs trial and error to come up with a solution to the problem.

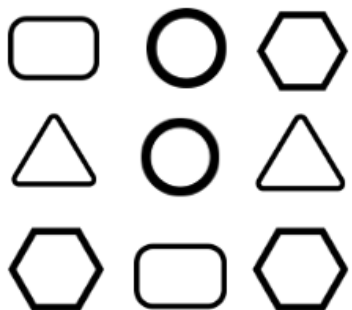
Source: www.deepsense.ai



Source: www.enjoyalgorithms.com

Deep learning. Unsupervised Learning

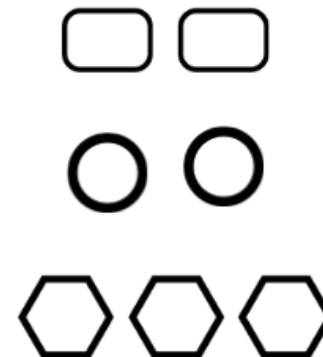
Unlabelled Data



Machine



Results



Unsupervised. Clean, perfectly labeled datasets aren't easy to come by. And sometimes, researchers are asking the algorithm questions they don't know the answer to. That's where unsupervised learning comes in.

In unsupervised learning, a deep learning model is handed a dataset without explicit instructions on what to do with it.

Artificial General Intelligence (AGI)

AGI is a machine with general intelligence much like a human being to solve any problem as humans do in any given situation.

AGI characteristics (those we see in all humans):

- Common Sense
- Background Knowledge
- Transfer Learning
- Abstraction
- Causality



Point of view A: a man apparently attacking another man



Point of view B: a man saving another man from an accident

AI researchers and scientists need to find a way to make machines conscious, programming a full set of cognitive abilities.

- Fujitsu-built K, built one of the fastest supercomputers also a notable attempt at achieving AGI, system took 40 minutes to simulate a single second of neural activity!!



Natural language processing, understanding and generation

NLG

- Discourse Generation
- Lexical Choice
- Sentence Planning/ Generation
- Realization
- Document Structuring

NLP

- Named entity Recognition (NER)
- Part-of-Speech Tagging (POS)
- Syntactic Parsing
- Coreference Resolution
- Machine Translation

NLU

- Lexical Ambiguity/ Analysis
- Sentiment Analysis
- Topic Classification
- Information Extraction
- Entity Detection
- Summarization
- Semantic Parsing
- Syntactical Analysis

Natural language processing, understanding and generation

NLP vs NLU vs. NLG

- **Natural language processing (NLP)** seeks to convert unstructured language data into a structured data format to enable machines to understand speech and text and formulate relevant, contextual responses. Its subtopics include natural language processing and natural language generation.
- **Natural language understanding (NLU)** focuses on machine reading comprehension through grammar and context, enabling it to determine the intended meaning of a sentence.
- **Natural language generation (NLG)** focuses on text generation, or the construction of text in English or other languages, by a machine and based on a given dataset.

What is a chatbot?

A **chatbot** is an artificial intelligence (AI) software that can simulate a conversation (or a chat) with a user in natural language through messaging applications, websites, mobile apps or through the telephone.

Why are chatbots important? A chatbot is often described as one of the most advanced and promising expressions of interaction between humans and machines. However, from a technological point of view, a chatbot only represents the natural evolution of a Question Answering system leveraging Natural Language Processing (NLP). Formulating responses to questions in natural language is one of the most typical Examples of Natural Language Processing applied in various enterprises' end-use applications.



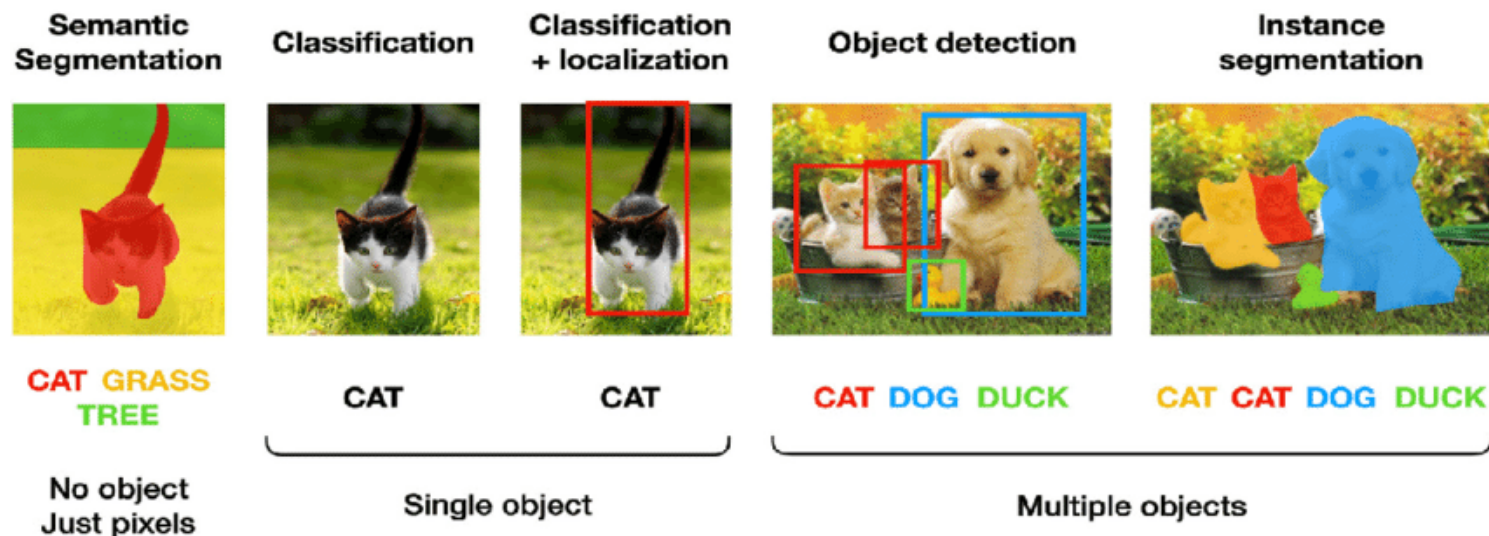
Source: www.experts.ai

What's computer vision?

Computer vision is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs — and take actions or make recommendations based on that information. If AI enables computers to think, computer vision enables them to see, observe and understand.

Computer vision works much the same as human vision, except humans have a head start. Human sight has the advantage of lifetimes of context to train how to tell objects apart, how far away they are, whether they are moving and whether there is something wrong in an image.

Source: IBM

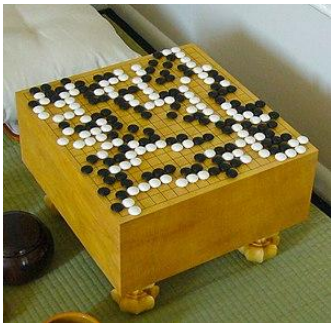


What's computer vision?

Deep Blue was an IBM supercomputer to play chess.

It was the first time a machine won Gary Kasparov on 10 February 1996 (Kasparov won the match!)

Deeper Blue won the match with Kasparov in May 1997.



AlphaGo program (from DeepMind, a Google company) beat the world champion professional Go player Lee Sedol in a five-game match in 2016.

AlphaZero beat the most powerful programs playing go, chess and shogi (Japanese chess) after a few days of play against itself using reinforcement learning.

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Source: Futures Platform, July 2018

Ethics Guidelines for Trustworthy AI

Ensure that the development, deployment and use of AI systems meets the seven key requirements for **Trustworthy AI**:

- Human agency and oversight
- Technical robustness and safety
- Privacy and data governance
- Transparency
- Diversity, non-discrimination and fairness
- Environmental and societal well-being
- Accountability.



Algoocracy

CHINA'S SOCIAL CREDIT SYSTEM

It's been dubbed the most ambitious experiment in digital social control ever undertaken. The Chinese government plans to launch its Social Credit System nationally by 2020.

WHAT'S THE AIM?

The system intends to monitor, rate and regulate the financial, social, moral and, possibly, political behavior of China's citizens – and also the country's companies – via a system of punishments and rewards. The stated aim is to "provide the trustworthy with benefits and discipline the untrustworthy."

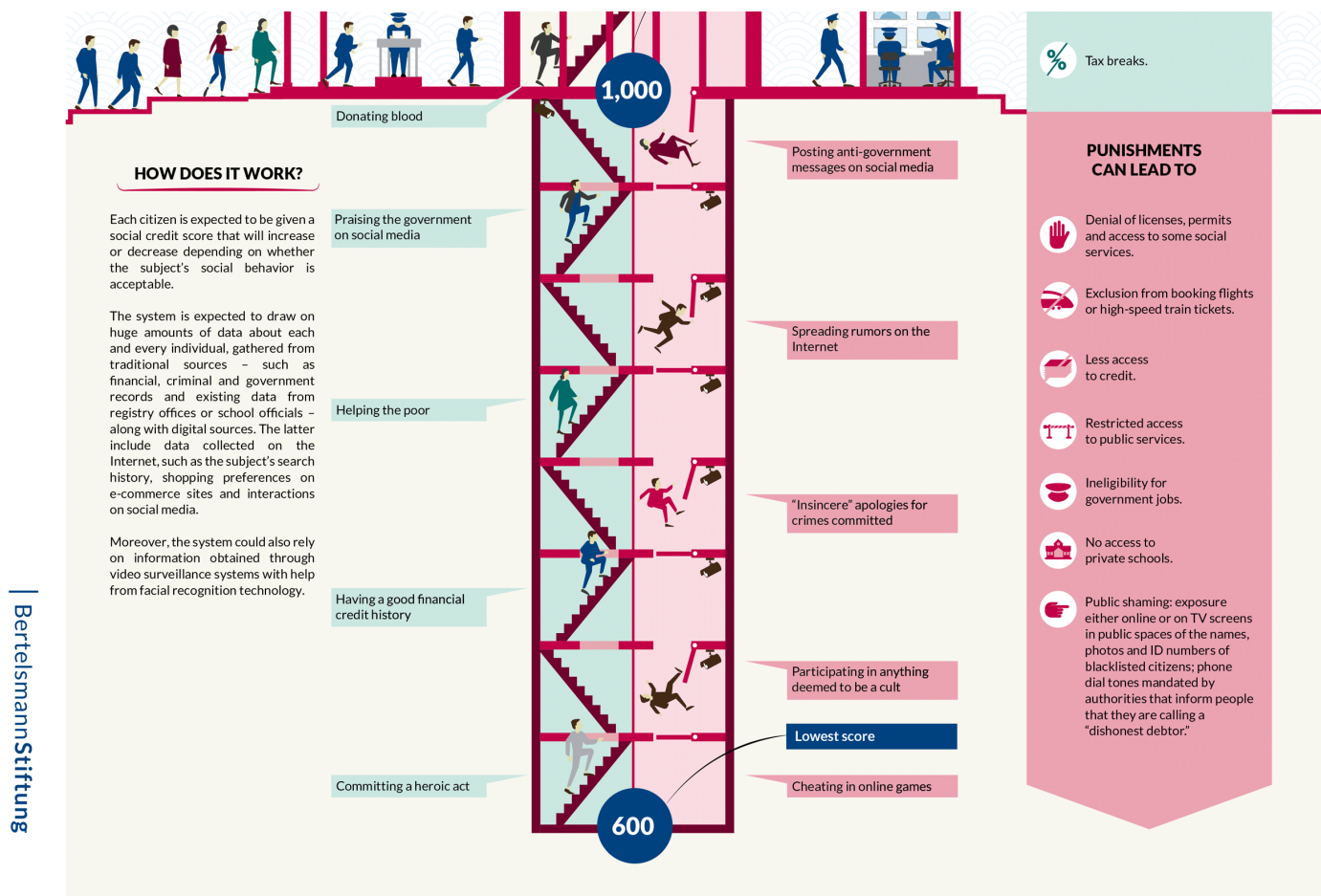
The Chinese government considers the system an important tool to steer China's economy and to govern society. There is still much speculation about how the final system will actually function. Details in this chart are based on pilot schemes and plausible expert expectations.

HOW DOES IT WORK?

TOP: BERHARD BARTSCH, MANTINGOTTRE, INFOGRAPHICS; CHRISTIAN EISENBERG, INFOGRAPHICS GROUP;
SOURCE: MERRICK, THE GUARDIAN, WIRED, DEUTSCHE WELLE



Algoocracy



The AI Debate: Development of Super intelligent general AI

The Beneficial AI movement (Techno-Skeptics)

- Super intelligent AI will ever come to pass (or in the remote future, too far)
- Regulation could impede AI developments
- AI could solve most problems, cures diseases, extends life indefinitely and frees us from drudgery
- AI systems must be robust and bug-free



Digital Utopians

- Regulation to limit AI developments to avoid doomsday scenarios
- Could we lose the ability to think for our self? Could we be denied the adventures of discovery and creativity? Could we be devalued, indoctrinated and lose freedoms, choices, emotions and experiences without even knowing it? Could we lose fundamental human rights, including the right of self-determination, the right to work, to self-care, privacy and basic human dignity? What price utopia?

Artificial Intelligence



Artificial Intelligence. Deep fake



AI + Robotics. Sophia



Artificial Intelligence + Robotics



PRACTICE: The AI Ethics Debate

- **Q1 Health:** Is it ethical to apply AI in health? Can we save lives? Can we generate viruses that attack only part of the population? Could we create beings without negative emotions?
- **Q2 Society:** Should AI results apply to the whole population or only to those who can afford them?
- **Q3 Geopolitics:** What can happen if legislation on AI between countries (or regions, in the case of the EU) is asymmetrical?



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